Final Project 1
Brain-to-Image Retrieval & Reconstruction

Overview

In this final project, your team will build and evaluate models that map **EEG brain signals** to **visual representations**.

You must complete **both** tasks below:

1. **Brain-to-Image Retrieval**
   Given an EEG segment, rank the correct stimulus image among a candidate set.
2. **Brain-to-Image Reconstruction**
   Given an EEG segment, generate an image that is consistent with the viewed stimulus.

The goal is **not** to reproduce any specific paper. You are expected to design and justify your own method under a common dataset and evaluation protocol.

Dataset

The dataset for this project is fixed, and for this project you only need to use data from **one designated subject** specified by the teaching team.

The data are collected from experiments in which a human subject views rapidly presented flickering images while the corresponding **EEG signals** are recorded. The resulting dataset pairs visual stimuli with brain responses and is designed for EEG-to-image retrieval and reconstruction tasks.

The TA team will provide:

- a fixed train/test split for the designated subject,
- preprocessed EEG data and/or preprocessing scripts,
- stimulus images and/or cached visual features,
- starter code,
- official evaluation scripts.

Data Format and Loading Notes

EEG signals are divided into multiple channels according to the placement of electrodes on different positions of the human scalp. The dataset provided in this project has already been preprocessed, including filtering and related standard preprocessing steps.

Each EEG sample has:

- **63 channels**,
- **250 time steps** on the temporal axis.

We also provide the channel name to channel index mapping file:

- `image-eeg-data_EEG_CHANNELS.jsonl`

You may choose to load all channels or only a selected subset of channels according to your method design.

Repeated Trials and `avg_trials`

In the training split, each experiment is repeated **80 times**. In the test split, each experiment is repeated **4 times**.

To improve the signal-to-noise ratio, it is **recommended** to set the `avg_trials` parameter to **true** during training. However, doing so will reduce the number of training samples to approximately **one eightieth** of the original training set size.

During testing, `avg_trials` **must** be set to **true**.

Rules

You may **not**:

- replace the dataset,
- use subjects other than the designated subject,
- modify the official train/test split,
- alter the logic of the TA evaluation scripts,
- use the test set for manual tuning.

Any additional data, pretrained models, or external resources must comply with course policy and must be clearly documented.

Task 1: Brain-to-Image Retrieval

Goal

From an EEG segment, produce a representation that can be matched against candidate images and used to rank them.

Evaluation

Retrieval will be evaluated with the official **200-way protocol** on the designated subject.

You must report:

- **Top-1 Accuracy**
- **Top-5 Accuracy**

Task 2: Brain-to-Image Reconstruction

Goal

From an EEG segment, generate an image that is structurally and semantically consistent with the viewed stimulus.

Evaluation

Reconstruction will be graded using the official TA evaluation code on the designated subject. Your report must include at least:

- **SSIM**
- **CLIP Score**

Additional perceptual or semantic metrics may also be used by the TA team.

Qualitative Results

Your report must include **8–12 qualitative examples**, each showing:

- the ground-truth stimulus image,
- your reconstructed image,
- a short discussion of success or failure cases.

Submission Requirements

Each team must submit:

1. Code

A runnable codebase including:

- data loading / preprocessing,
- model definitions,
- training scripts,
- inference scripts,
- output generation for official evaluation.

2. Technical Report

Your report should include:

- problem formulation,
- method description,
- training and inference procedure,
- experimental setup,
- quantitative results,
- qualitative results,
- analysis and discussion,
- disclosure of any external resources used.

3. Prediction Outputs

Submit valid output files for:

- retrieval predictions,
- reconstruction results,

in the format required by the official TA scripts.

Grading

Total: **100 points**

- **Retrieval leaderboard performance**: 25 pts
  - Top-1 ranking: 12.5 pts
  - Top-5 ranking: 12.5 pts

- **Reconstruction leaderboard performance**: 25 pts
  - CLIP Score ranking: 12.5 pts
  - SSIM ranking: 12.5 pts
- **Technical design and methodology**: 20 pts
- **Experiments and report quality**: 20 pts
- **Code quality and reproducibility**: 10 pts

Important

Both tasks are **mandatory**.

If your submission only completes one task, or fails to produce valid outputs for official evaluation in one task, it will be considered incomplete. Unless otherwise announced, incomplete submissions may receive **at most 60/100**.

Reproducibility is a basic requirement of this project. If the TA team cannot run your code successfully and reproduce the submitted results, the submission will be treated as **non-reproducible** and will receive **0 points**.

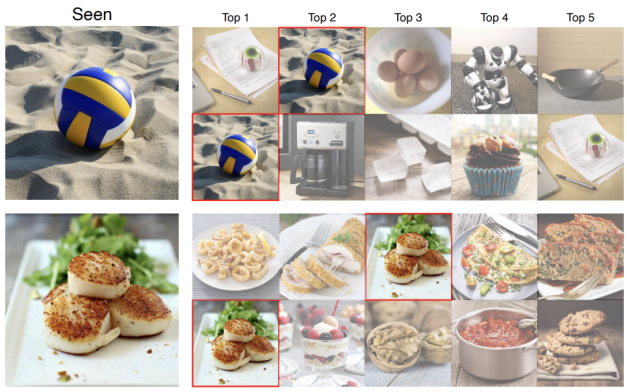

Figure 1: Retrieval

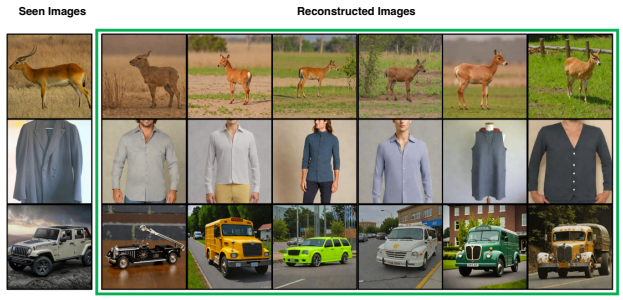

Figure 2: Recon

Academic Integrity

You may collaborate only within your own team.

The following are prohibited:

- copying code or reports from other teams,
- fabricating results,
- altering evaluation outputs,
- using unauthorized external data or test information,
- failing to disclose external resources.

Any violation will be handled according to course academic integrity policy.

Summary

Your team must build a complete EEG-to-image system on the provided dataset and evaluate it from two perspectives:

- **retrieval quality**, and
- **reconstruction quality**.

A strong project should demonstrate not only good performance, but also sound methodology, careful experimentation, clear analysis, and full reproducibility.